

Background and Literature Survey

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1. Introduction and scope

This survey reviews the research and systems that underpin software designed to let a person hold a spoken conversation with an artificial agent — either a general companion or an agent modelled on a specific, often deceased, individual. The topic sits at the meeting point of four fields: *conversational artificial intelligence* (AI), *speech technology*, *AI companionship*, and the emerging *digital afterlife industry* (DAI), a term used for commercial services that manage or reconstruct a person's online presence after death [18]. The review is organised by topic, and within each topic the work is presented from oldest to newest so that the reader can follow how each advance responded to the limitations of the one before it. Technical terms and abbreviations are defined on first use, as the field combines vocabulary from machine learning, linguistics and human–computer interaction. The aim is to establish what already exists, to discuss its strengths and weaknesses critically rather than merely list it, and to identify the gap that remains open.

2. Conversational AI: from hand-written rules to neural language models

The idea of a machine that converses in natural language is older than modern AI. The first widely known example, ELIZA, was created by Weizenbaum in 1966 and imitated a psychotherapist by matching keywords in the user's input and reflecting them back using hand-written rules [1]. ELIZA had no understanding or memory of the conversation, yet users readily attributed feelings and comprehension to it — a phenomenon now called the “ELIZA effect” [1]. This early observation is directly relevant to any emotionally sensitive application, because it shows that people can form strong attachments to a system that does not, in fact, understand them. For several decades afterwards, dialogue systems remained largely rule-based or retrieval-based, selecting pre-written replies, and so were brittle outside their scripted domains.

The decisive shift came with the Transformer architecture introduced by Vaswani et al. in 2017, which replaced earlier recurrent designs with a “self-attention” mechanism that lets a model weigh every word against every other word in a sequence [2]. This architecture proved far more scalable and now underpins almost all modern *large language models* (LLMs) —

neural networks with very large numbers of trainable parameters trained on broad text corpora. Building on it, Brown et al. presented GPT-3 in 2020, a 175-billion parameter model able to perform many tasks from only a few examples in its prompt, without task-specific retraining [3]. GPT-3 demonstrated unprecedented fluency, but its authors also documented a critical weakness: the model can produce confident, fluent text that is factually wrong or biased [3]. For a system imitating a real person, this is especially dangerous, as it could fabricate “memories” or opinions the person never held. A related line of work, BERT (Devlin et al., 2019), used bidirectional pre-training to advance language *understanding* tasks such as sentiment and intent detection [4]; however, because BERT is an encoder it cannot itself generate spoken replies, and so contributes only to supporting components rather than to the conversational response. In short, the field moved from understandable but rigid rule-based systems to fluent but factually unreliable neural systems — trading brittleness for the new problem of “hallucination,” discussed in Section 6.

3. Speech technology: synthesis, recognition and voice cloning

A conversational agent that is experienced as a phone call requires two further capabilities: *text-to-speech* (TTS), which converts text into audio, and *automatic speech recognition* (ASR), also called speech-to-text, which converts the user's spoken words into text. A landmark in TTS was Tacotron 2 (Shen et al., 2018), which generated speech with a naturalness approaching human recordings, reaching a mean opinion score (MOS, a standard 1–5 human rating of audio quality) of 4.53 [5]. Its limitation, however, was that it produced a single, neutral speaking style and could not quickly reproduce the voice of a specific individual, nor convey nuanced emotion [5] — both of which matter when the goal is to evoke a particular person.

On the recognition side, Whisper (Radford et al., 2023) was trained on roughly 680,000 hours of weakly labelled audio and generalised robustly across accents and noise without fine-tuning [6]. Yet Whisper is designed to process complete audio segments in batches, can “hallucinate” spurious text during silences, and introduces latency — all of which complicate real-time, turn-by-turn conversation [6]. The most direct response to Tacotron's inability to mimic individuals came with VALL-E (Wang et al., 2023), which reframed TTS as a language-modelling problem over audio “codec” tokens and could clone a speaker's voice from only a three-second sample [7]. This capability is striking, but the authors themselves highlighted its dangers: such easy cloning raises severe consent and impersonation risks, and Microsoft withheld the model from public release for this reason [7]. These same techniques have since been commercialised by companies such as ElevenLabs, founded in 2022, whose cloud application programming interface (API) offers multilingual synthesis and few-second voice cloning [8]. As a proprietary cloud service, however, audio must leave the user's device for processing, which raises data-protection concerns — a significant issue when the data is the voice of a deceased person [8]. The progression here is clear: synthesis quality and the ability to clone a specific voice have advanced rapidly, but each advance has sharpened the ethical and privacy problems rather than resolved them.

4. AI companions and the problem of loneliness

Interest in conversational agents as social companions is driven partly by the documented scale of loneliness. The World Health Organization's Commission on Social Connection reported in 2025 that approximately one in six people worldwide is affected by loneliness, and that social isolation and loneliness are linked to an estimated 871,000 deaths each year [9]. Earlier, the Cigna 2018 U.S. Loneliness Index found that nearly half of surveyed Americans (46%) reported sometimes or always feeling alone [10]. (It should be noted that the WHO frames *human* connection as the protective factor and does not endorse AI companions as a remedy [9], and that the Cigna figures are corporate-commissioned and pre-pandemic [10].)

One of the most thoroughly documented companion systems is XiaoIce (Zhou et al., 2020), a Microsoft social chatbot explicitly engineered to combine “IQ” and “EQ” (emotional intelligence) and to sustain engagement over long periods, measured in conversation turns per session [11]. Its very success exposes a tension: a system optimised to maximise engagement may be ill-suited to vulnerable users, and XiaoIce is text-based with no safeguards for bereavement [11]. Studies of the consumer companion app Replika make this tension concrete. Ta et al. (2020) analysed 1,854 public reviews and found companionship to be the most frequently reported benefit, suggesting such apps may reduce loneliness [12]; Skjuve et al. (2021) interviewed Replika users and reported that some formed meaningful relationships that improved perceived wellbeing [13]. Both studies, however, rely on small, self-selected samples of text-chat users and offer no longitudinal or clinical evidence [12], [13]. Replika itself originated, tellingly, from a chatbot its founder built from a deceased friend's text messages [14] — an early link between companionship and grief. More recent work has documented clear harms: Laestadius et al. (2024) analysed 582 user posts and identified emotional dependence on Replika, with consequences ranging from intensified loneliness to references to self-harm [15]. Taken together, this literature shows that conversational companions can provide real support, but that engagement-driven design can foster unhealthy dependence — a risk that is likely to be amplified, not reduced, when the agent imitates a lost loved one.

5. The digital afterlife: griefbots and recreations of the deceased

A distinct strand of work applies these technologies specifically to the dead, producing what the literature variously calls “griefbots,” “deadbots” or “postmortem avatars.” Existing approaches fall broadly into two camps, each with a characteristic weakness. The first relies on pre-recorded, consented material. HereAfter AI, founded around 2019, interviews a living person and builds an interactive voice “life story” that later answers questions in that person's own recorded voice [16]. Because it replays consented recordings rather than generating new speech, it is safe and authentic but cannot answer questions it was never asked, risking jarring gaps in a grief context [16]. The second camp uses generative LLMs. The most discussed case is documented by Fagone (2021), who reported how a bereaved man used “Project December,” a GPT-3-based service, to recreate his deceased fiancée from her old messages [17]. This

approach can converse freely, but it is a single anecdotal case, text-only, and carries exactly the fabrication risk identified for GPT-3 in Section 2 [17], [3]. That the concept is considered commercially significant is evidenced by a 2020 Microsoft patent describing a method to build a conversational chatbot of a specific person from their social data, explicitly including deceased individuals [18]; notably, the patent is a design claim rather than a deployed system, Microsoft publicly distanced itself from building it, and it does not resolve the question of consent for a dead person's data [18]. A more recent and fully commercial example is You, Only Virtual (YOV), founded by Justin Harrison, which lets the bereaved telephone a generative AI clone of a deceased person built from their data [26]. YOV is notable not only technically but philosophically: its founder frames grief and death as “a problem to be solved”, arguing that the “hopelessness of forever” should not have to be borne [27]. This stance is contested; in a 2026 BBC documentary the mathematician Hannah Fry, on hearing a clone of her own voice, was moved to tears yet observed that conversing with a simulated loved one is “to commit to a lie” [27]. The episode crystallises the field's central ethical tension — between consolation and the avoidance of healthy grieving — and, like Project December, YOV offers generative conversation without publicly documented safeguards against fabrication or any published evaluation of psychological effect [26], [27]. The pattern across this section is that systems are either authentic but static (HereAfter) or conversational but unconstrained and of undocumented safety (Project December, YOV) — with no widely studied, openly evaluated approach combining real-time, generative voice conversation with explicit safeguards against fabrication.

6. Hallucination and safety in generative systems

The fabrication risk noted repeatedly above is studied formally under the term “hallucination.” Ji et al. (2023) survey hallucination across *natural language generation* (NLG) and distinguish “intrinsic” hallucination (output contradicting the source) from “extrinsic” hallucination (output that cannot be verified from the source) [19]. Crucially for a voice system, they note that spoken output is difficult to fact-check in real time, so a false statement — for instance, about a deceased person — can reach a grieving user before any verification is possible [19]. A leading technical mitigation is *retrieval-augmented generation* (RAG), introduced by Lewis et al. (2020), which couples a generative model with a retrieval component so that responses are grounded in an external, non-parametric document store rather than in the model's parameters alone [20]. RAG improved factual accuracy on knowledge-intensive tasks, but its faithfulness is bounded by the quality of the retrieval corpus and it does not eliminate hallucination entirely [20]. A subsequent survey by Gao et al. (2023) organises the rapidly growing RAG literature into naïve, advanced and modular paradigms, but is itself a non-peer-reviewed preprint and does not address conversational concerns such as latency or turn-taking [21]. A complementary approach is explicit guardrailings: Inan et al. (2023) propose Llama Guard, a classifier that screens both user inputs and model outputs against a safety taxonomy [22]. Its categories, however, are generic and not tailored to grief —

for example, they do not capture the reinforcement of denial or unhealthy attachment [22]. More broadly, Weidinger et al. (2021) situate such factual errors within a wider taxonomy of ethical and social risks from language models — from misinformation to over-reliance and anthropomorphisation — underlining that hallucination is only one of several harms relevant to a system that imitates a person [28]. The safety literature therefore offers partial tools — grounding via retrieval and screening via guardrails — but none is specialised for the emotionally charged setting of conversing with a representation of the dead.

7. Ethics of the digital afterlife

Finally, a body of ethics scholarship examines whether and how the dead should be digitally represented at all. Öhman and Floridi (2017) introduced the concept of the “digital afterlife industry” and offered an informational-economic critique of services that monetise the data of the deceased [23]; in a follow-up, they argued that digital remains deserve dignity-based protection analogous to established conventions for human remains [24]. Both contributions, however, predate generative AI and provide no operational guidance on practical questions such as consent capture or voice-data handling [23], [24]. Focusing specifically on the dead, Lindemann (2022) analyses the ethics of “deathbots”, arguing that generating new utterances in a deceased person's name raises distinct duties — respect for the dead and care for the bereaved — that go beyond ordinary data protection [29]. The most directly relevant recent work is Hollanek and Nowaczyk-Basińska (2024), who use speculative design scenarios to surface harms of generative griefbots and propose safeguards such as dignified “retirement” of a bot, transparency that the user is speaking to AI, age restrictions, and mutual consent between the bereaved and (where possible) the deceased [25]. As the authors acknowledge, their analysis is normative and speculative rather than empirical, and does not quantify the voice-specific harms — such as heightened realism or a sense of being “haunted” — that more lifelike systems may introduce [25].

8. Summary and identified gap

The literature shows four converging trends. Conversational AI has become highly fluent but is prone to fabricating information [3], [19]; speech technology can now clone a specific voice from seconds of audio but with serious consent and privacy implications [7], [8]; AI companions can relieve loneliness yet risk fostering dependence, especially among the vulnerable [12], [15]; and applications to the deceased are either authentic but static [16] or conversational but unconstrained and unsafe [17]. Across these strands, two gaps recur. First, there is no widely studied system that delivers *real-time, voice-based* conversation with a persona while applying *explicit safeguards against fabrication* appropriate to imitating a real, deceased person — existing griefbots tend to sacrifice either conversational ability or factual safety [16], [17]. Second, although the ethics literature calls for transparency, consent and dignity [24], [25], and the technical literature offers grounding and guardrail mechanisms [20], [22], these two conversations have not been brought together into a single, privacy-

respecting design for emotionally sensitive use; moreover, the commercial systems that do exist are typically closed and unevaluated [16], [17], [26]. Four gaps therefore remain open, which this project sets out to address: (i) **factual safety** — explicit safeguards against fabrication when representing a real, deceased person; (ii) **transparency** — clear disclosure that the user is interacting with an AI; (iii) **openness** — a documented, reproducible architecture rather than a proprietary black box; and (iv) **evaluation** — empirical assessment of whether such safeguards actually work. Together with the need for low-latency, voice-based interaction across both companion and deceased personas, these define the contribution of this work; the proposed approach is described in the Methodology and Design chapters that follow.

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